Note

In a finite A.P. whose first term is a and last term l, then the number of terms in the A.P. is given by $l = a + (n-1)d \Rightarrow n = \left(\frac{l-a}{d}\right) + 1$

Example 2.26 Find the number of terms in the A.P. 3, 6, 9, 12,..., 111.

Solution

First term $a = 3$; common difference $d = 6 - 3 = 3$; last term $l = 111$

We know that, $n = \left(\frac{l-a}{d}\right) + 1$

$$n = \left(\frac{111-3}{3}\right) + 1 = 37$$

Thus the A.P. contain 37 terms.

Progress Check

1. The common difference of a constant A.P. is _____.
2. If a and l are first and last terms of an A.P. then the number of terms is _____.

Example 2.27 Determine the general term of an A.P. whose 7th term is -1 and 16th term is 17.

Solution Let the A.P. be $t_1, t_2, t_3, t_4, \ldots$

It is given that $t_7 = -1$ and $t_{16} = 17$

$a + (7-1)d = -1$ and $a + (16-1)d = 17$

$$a + 6d = -1 \quad \ldots(1)$$

$$a + 15d = 17 \quad \ldots(2)$$

Subtracting equation (1) from equation (2), we get $9d = 18 \Rightarrow d = 2$

Putting $d = 2$ in equation (1), we get $a + 12 = -1 \therefore a = -13$

Hence, general term $t_n = a + (n-1)d$

$$= -13 + (n-1) \times 2 = 2n - 15$$

Example 2.28 If l^{th}, m^{th} and n^{th} terms of an A.P. are x, y, z respectively, then show that

(i) $x(m-n) + y(n-l) + z(l-m) = 0$ (ii) $(x-y)n + (y-z)l + (z-x)m = 0$

Solution (i) Let a be the first term and d be the common difference. It is given that

$t_l = x, t_m = y, t_n = z$

Using the general term formula

$$a + (l-1)d = x \quad \ldots(1)$$

$$a + (m-1)d = y \quad \ldots(2)$$

$$a + (n-1)d = z \quad \ldots(3)$$

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